Multimodal Disease Severity Assessment & Temporal Progression Forecasting

For Cavendish Banana Plantations — Component 3

COMPREHENSIVE REVISED TECHNICAL STUDY GUIDE

Prepared for University Viva and Academic Defense

Scope: Complete technical explanation of the 8-stage inference pipeline, mathematical foundations, implementation architecture, experimental results, known limitations, and 30 prepared viva defense questions.

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Chapter 1: Introduction and Research Context

1.1 Research Motivation

Cavendish banana (*Musa* AAA group) is the world's dominant dessert banana cultivar, yet it is increasingly threatened by foliar diseases — Black Sigatoka (*Mycosphaerella fijiensis*), Yellow Sigatoka (*Mycosphaerella musicola*), and Panama Disease / Fusarium Wilt (*Fusarium oxysporum* f. sp. *cubense*). Traditional assessment relies on trained agronomists who visually estimate disease severity using ordinal scales (e.g., Stover's Modified Scale). This manual approach has well-documented limitations: inter-rater agreement between trained pathologists is typically only ~71%, assessments are time-consuming, and they cannot be scaled across large plantation estates.

Component 3 addresses these challenges by building a fully automated, **multimodal severity assessment and temporal progression forecasting** pipeline. The system ingests a single RGB leaf image and produces: (i) pixel-level segmentation of healthy tissue vs. diseased lesions, (ii) morphometric feature vectors describing lesion geometry and leaf shape, (iii) disease-specific physiological impact scores, (iv) fused severity estimates via multiple analytical strategies, (v) calibrated XGBoost regression predictions, and (vi) longitudinal trend classification with GRU-based future severity forecasting.

1.2 Formal Problem Statement

Given an RGB banana leaf image I of shape (H, W, 3), produce a real-valued composite severity score S in [0, 100] that integrates:

- **Visual disease coverage** from pixel-level semantic segmentation
- **Morphometric characteristics** of lesion spatial distribution and leaf shape
- **Disease-specific physiological impact** derived from domain plant pathology
- **Temporal trajectories** of severity over repeated observations of the same plant

$$f: I \in \mathbb{R}^{H \times W \times 3} \rightarrow S \in [0, 100]$$

The end-to-end mapping from raw RGB image to composite severity percentage.

1.3 Why Multimodal Fusion?

No single signal is sufficient for accurate severity assessment. A leaf with 12% lesion coverage where the largest lesion coalesces across 9% of the lamina is clinically more severe than one with 12% coverage distributed across 20 tiny spots. Similarly, a leaf exhibiting morphological stunting (low aspect ratio) alongside lesions indicates systemic disease impact beyond surface symptoms. Our multimodal pipeline therefore combines pixel-level, geometric, and pathological signals to produce a richer severity estimate than any unimodal approach.

1.4 Three-Component Architecture Overview

This project is part of a three-component system. **Component 1** handles disease classification (identifying which disease is present). **Component 2** addresses symptom detection and localization. **Component 3** (this work) quantifies *how severe* the disease is and *how it progresses* over time. Component 3 receives the disease class label from Component 1 and optionally uses localization data from Component 2, but its core analysis pipeline operates independently on the input leaf image.

VIVA INSIGHT: DISTINGUISHING COMPONENTS

In a viva, you may be asked: "How does Component 3 differ from Components 1 and 2?" The answer is that C1 classifies disease *type* (categorical), C2 detects symptom *location* (spatial), and C3 quantifies disease *severity* (continuous regression) and *temporal trajectory* (time-series forecasting). C3 is the only component that produces a numeric severity score and performs longitudinal analysis.

1.5 Ground-Truth Strategy and Known Circularity

A critical methodological point concerns our ground-truth labels. The AgroExpertSimulator class (src/evaluation/expert_simulator.py) generates "expert ratings" by injecting Gaussian noise ($\sigma = 4.2$ percentage points) around the pipeline's own fusion output F_{hybrid}. This noise level was calibrated to match the approximately 71% inter-rater agreement reported in plant pathology literature for manual visual assessment.

LIMITATION / CAUTION: CIRCULARITY IN EVALUATION

This is **not** independent ground truth. The simulated "expert" ratings are mathematically derived from the system's own predictions. This means high correlation metrics (e.g., Spearman $\rho \approx 0.99$) between predictions and "ground truth" are a tautological artifact of this design. In a real deployment, one would validate against actual agronomist ratings or calibrated disease severity scales (e.g., Stover's Modified Scale). We acknowledge this circularity explicitly. The purpose of AgroExpertSimulator is to model the *statistical distribution* of human assessment noise, not to serve as an independent validation oracle.

$$\text{Expert_rating} = \text{clip}(F_{hybrid} + N(0, \sigma^2), 0, 100) \text{ where } \sigma = 4.2$$

Each simulated expert adds independent Gaussian noise to the pipeline's own fusion output. Three experts form a panel whose consensus is their arithmetic mean.

1.6 Dataset and Data Generation

The project uses a synthetic longitudinal dataset generated by the data module (src/data/generator.py). The dataset comprises 120 simulated banana plants, each with 3-5 temporal observation points, yielding approximately 480 image-mask pairs. Each pair consists of a 256×256 RGB leaf image and a corresponding 3-class segmentation mask (background=0, healthy leaf=1, lesion=2). Lesion patterns are procedurally generated to simulate realistic disease progression with varying severity levels, spatial distributions, and coalescence patterns across the three target diseases.

Data partitioning uses **plant-level stratified splitting** to prevent data leakage: 70% training (84 plants), 15% validation (18 plants), 15% test (18 plants). No images from the same plant appear in multiple splits, ensuring the model cannot memorize temporal sequences during training.

Chapter 2: System Architecture and Pipeline Design

2.1 The 8-Stage Inference Pipeline

The Component3Orchestrator (src/pipeline/c3_orchestrator.py) executes a deterministic 8-stage sequential pipeline. Every stage receives the accumulated state from prior stages and appends its outputs to a shared result dictionary:

Stage	Name	Description
Stage 1	Image Preprocessing	Resize to 256×256, normalize to [0,1], convert to NCHW PyTorch tensor
Stage 2	U-Net Segmentation	3-class pixel prediction: background (0), healthy leaf (1), lesion (2)
Stage 3	Morphometric Extraction	Compute scope%, lesion count, coalescence ratio, aspect ratio, distortion index
Stage 4	Impact Scoring	Disease-specific physiological severity computation using pathology-informed formulas
Stage 5	Multimodal Fusion	Six parallel fusion strategies producing diverse severity estimates
Stage 6	XGBoost Regression	Machine-learned severity prediction from the 5D morphometric feature vector
Stage 7	Temporal Persistence	Store observation in SQLite; compute OLS trend across plant history
Stage 8	GRU Forecasting	Predict next-step severity using recurrent neural network on temporal sequence

2.2 Data Flow Architecture

The pipeline is strictly feed-forward at inference time. Data flows through the stages as follows:

- **Input:** Raw RGB image (any resolution) + disease_class string + plant_id + timestamp
- **Stage 1→2:** Preprocessed tensor (1, 3, 256, 256) → U-Net → segmentation mask (256, 256)
- **Stage 2→3:** Segmentation mask → MorphometricFeatures dataclass (5 numeric fields)
- **Stage 3→4:** MorphometricFeatures + disease_class → DiseaseImpactScore float
- **Stage 3+4→5:** (MorphometricFeatures, ImpactScore, disease_class) → 6 fusion scores
- **Stage 3→6:** 5D feature vector [scope, lesions, coalescence, AR, distortion] → XGBoost scalar
- **Stage 5→7:** Selected fusion score → SQLite row → OLS trend {improving, stable, worsening} + slope
- **Stage 7→8:** Historical severity sequence → GRU → next-step prediction ± confidence interval

2.3 Technology Stack

Component	Technology	Purpose
Deep Learning	PyTorch 2.x	U-Net segmentation + GRU forecasting
ML Regression	XGBoost (scikit-learn API)	Severity regression from morphometrics
Computer Vision	OpenCV (cv2)	Connected component analysis, contours, morphometry
API Layer	FastAPI + Uvicorn	REST endpoints for inference and visualization
Database	SQLite3	Temporal observation persistence and retrieval
Statistics	SciPy + NumPy	OLS regression, correlation metrics, distribution fitting
Frontend	HTML/CSS/JS (SPA)	Interactive dashboard with dual visualization panels

2.4 Design Rationale: Why Sequential over Parallel?

The pipeline is deliberately sequential because each stage depends on prior outputs. U-Net must produce the mask before morphometrics can be computed; impact scoring requires both the morphometric features and the disease class; fusion requires all prior signals. Only within Stage 5 do we compute six strategies in parallel (they share the same inputs). This sequential design also enables deterministic debugging — each stage's output is logged and can be inspected independently via the API's /analyze endpoint.

VIVA INSIGHT: PIPELINE ORDERING

In a viva you might be asked: "Why not run segmentation and feature extraction in parallel?" The answer is that feature extraction requires the segmentation mask as input — these stages are inherently sequential due to data dependencies. Only independent computations like the six fusion strategies can run in parallel.

2.5 Configuration Management

All hyperparameters are centralized in config/default_config.yaml and accessed via src/utils/config.py. This includes U-Net architecture parameters (base_filters, input size), training hyperparameters (learning rate, epochs, batch size), XGBoost regression parameters (n_estimators, max_depth), GRU parameters (hidden_dim, num_layers), and fusion strategy weights. This centralization allows reproducible experiments and systematic ablation studies.

Chapter 3: U-Net Semantic Segmentation

3.1 Why U-Net for Foliar Segmentation?

U-Net was originally designed for biomedical image segmentation (Ronneberger et al., 2015) and has become the standard architecture for pixel-level classification tasks where training data is limited. Its key innovation is the **encoder-decoder architecture with skip connections**, which concatenates spatially rich encoder feature maps with the upsampled decoder feature maps. This allows the network to combine both *semantic context* (from deep layers) and *spatial precision* (from shallow layers) — critical for accurately delineating irregular lesion boundaries.

We use a **3-class U-Net** (background=0, healthy leaf=1, lesion=2) with base_filters=64 (configurable, default 32 in training for faster convergence). The architecture follows the canonical U-Net topology: 4 encoding levels → bottleneck → 4 decoding levels, each with skip connections.

3.2 Architecture Implementation Details

The implementation (src/segmentation/unet.py) defines three building blocks:

DoubleConv: Each level applies two consecutive 3×3 convolutions, each followed by BatchNorm2d and ReLU activation. This double convolution extracts increasingly abstract features while BatchNorm stabilizes training gradients.

Down (Encoder): MaxPool2d(2×2) followed by DoubleConv. The max pooling halves spatial resolution while the convolutions double the channel count: 64→128→256→512→1024.

Up (Decoder): Transposed convolution (ConvTranspose2d, stride=2) or bilinear upsampling, followed by concatenation with the corresponding encoder skip connection, then DoubleConv. Channels decrease: 1024→512→256→128→64.

OutConv: A single 1×1 convolution that projects the final 64-channel feature map to n_classes=3 output channels (unnormalized logits).

Layer	Input Channels	Output Channels	Spatial Resolution	Operation
inc (input)	3	64	256×256	DoubleConv
down1	64	128	128×128	MaxPool + DoubleConv
down2	128	256	64×64	MaxPool + DoubleConv
down3	256	512	32×32	MaxPool + DoubleConv
down4 (bottleneck)	512	1024	16×16	MaxPool + DoubleConv
up1	1024+512	512	32×32	Upsample + Cat + DoubleConv
up2	512+256	256	64×64	Upsample + Cat + DoubleConv
up3	256+128	128	128×128	Upsample + Cat + DoubleConv
up4	128+64	64	256×256	Upsample + Cat + DoubleConv
outc	64	3	256×256	1×1 Conv (logits)

3.3 Combined Dice-CrossEntropy Loss

Our training loss is a weighted sum of two complementary objectives:

$$L_{total} = 0.5 \times L_{dice} + 0.5 \times L_{CE}$$

Equal weighting of Dice loss (overlap-based) and Cross-Entropy (pixel-wise classification).

Soft Dice Loss (as implemented in loss_and_metrics.py):

$$Dice_c = (2 \times \Sigma\ p_c \times y_c + \varepsilon) / (\Sigma\ p_c^2 + \Sigma\ y_c^2 + \varepsilon)$$

Where p_c = softmax probabilities for class c, y_c = one-hot ground truth for class c, ε = 1e-6 (Laplace smoothing). Note: our implementation uses $p^2 + y^2$ in the denominator (not $p + y$ as in standard Sørensen-Dice). This is the V-Net variant of Dice loss.

L_dice = 1 - mean(Dice_c) over all classes c

The final Dice loss is 1 minus the mean Dice coefficient across all 3 classes.

VIVA INSIGHT: DICE LOSS VARIANT

The denominator uses $p^2 + y^2$ (sum of squares) rather than the more common $|p| + |y|$ (sum of absolute values). This is a design choice from the V-Net paper (Milletari et al., 2016). Both formulations produce smooth, differentiable overlap measures, but the squared variant penalizes confident false predictions more heavily. You should be prepared to explain this distinction in a viva.

Cross-Entropy Loss:

L_CE = -(1/N) × Σ_i Σ_c y_ic × log(p_ic)

Standard pixel-wise multi-class cross-entropy. N = total pixels, C = 3 classes.

Why combine both? Cross-entropy treats each pixel independently and is effective for pixel-level accuracy. Dice loss operates on the *set overlap* between predicted and ground truth regions, making it robust to class imbalance (background pixels vastly outnumber lesion pixels). The combination gives strong per-pixel gradients (CE) while encouraging accurate region-level overlap (Dice).

3.4 Training Configuration

Hyperparameter	Value	Rationale
Optimizer	AdamW	Decoupled weight decay; improved generalization vs. Adam
Learning Rate	0.001 (initial)	Standard for U-Net; decayed via cosine schedule
LR Schedule	CosineAnnealingLR (T_max=epochs)	Smooth annealing to η_min=1e-5; avoids sharp drops
Weight Decay	1e-4	L2 regularization to prevent overfitting
Batch Size	8	Balances GPU memory with gradient noise for generalization
Epochs	10 (configurable)	With early stopping (patience=5) on val Macro Dice
Input Size	256×256 RGB	Standard resolution; balances detail with compute cost
Data Split	84/18/18 plants	Plant-level partitioning (no temporal data leakage)

3.5 Inference: predict_mask()

At inference time, the predict_mask() method bypasses loss computation:

1. Set model to eval mode (disabling dropout and using running BatchNorm statistics).
2. Forward pass produces logits tensor of shape (B, 3, H, W).
3. Apply argmax along the channel dimension: predicted_class = argmax(logits, dim=1).
4. The result is a (H, W) integer mask with values in {0, 1, 2}.

TECHNICAL NOTE: INFERENCE OPTIMIZATION

The predict_mask() method uses @torch.no_grad() decorator, which disables gradient computation and reduces memory usage during inference. This is critical for deployment efficiency.

3.6 Segmentation Evaluation Metrics

We evaluate segmentation quality using per-class and macro-averaged metrics:

IoU_c = TP / (TP + FP + FN + ε)

Intersection over Union for class c. TP=true positives, FP=false positives, FN=false negatives.

Dice_c = 2×TP / (2×TP + FP + FN + ε)

Discrete (hard) Dice coefficient for evaluation (different from the soft Dice used in the loss).

The evaluation also computes **Total Leaf metrics** by combining classes 1 and 2 (healthy leaf ∪ lesion = total leaf lamina), giving whole-leaf segmentation quality. Macro IoU and Macro Dice are arithmetic means across all 3 classes.

Chapter 4: Morphometric Feature Extraction

4.1 Purpose and Motivation

The U-Net produces a pixel-level mask, but raw pixel counts are insufficient for severity assessment. Two leaves with identical 15% lesion coverage may have very different disease severity: one with a single large coalesced lesion (high severity) and another with 30 scattered tiny spots (lower severity). The Morphometric Scope Engine (src/segmentation/scope_calculator.py) extracts a **5-dimensional feature vector** that captures coverage, spatial distribution, coalescence, and leaf shape — all from the 3-class segmentation mask.

4.2 The Five Morphometric Features

Feature 1: Scope Percentage (S)

$$S = (\text{lesion_pixels} / \text{leaf_pixels}) \times 100$$

Where leaf_pixels = pixels labeled 1 (healthy) + pixels labeled 2 (lesion). This is the total leaf lamina area. $S \in [0, 100]$, clipped.

This measures the proportion of the leaf lamina that shows visible disease symptoms. A leaf with 1000 healthy pixels and 200 lesion pixels has S = 200/1200 × 100 = 16.67%.

Feature 2: Lesion Count (N)

The number of distinct contiguous lesion regions, computed using OpenCV's connectedComponentsWithStats() with 8-connectivity. A minimum area threshold of **5 pixels** filters out single-pixel noise artifacts.

SOURCE: scope_calculator.py

```
lesion_uint8 = lesion_bool.astype(np.uint8) * 255
num_labels, labels, stats, _ = cv2.connectedComponentsWithStats(
    lesion_uint8, connectivity=8
)
# Count only components with area >= min_lesion_pixels (5)
```

Feature 3: Coalescence Ratio (R_coal)

$$R_coal = (\text{max_lesion_area} / \text{leaf_pixels}) \times 100$$

Ratio of the single largest contiguous lesion to total leaf area. High R_coal indicates lesions are merging (coalescing), a sign of advanced disease.

If the largest single lesion covers 150 of 1200 leaf pixels, R_coal = 150/1200 × 100 = 12.5%. Compare this with total scope S — if R_coal is close to S, nearly all disease is in one large patch. If R_coal is much smaller than S, disease is distributed across many small spots.

Feature 4: Leaf Aspect Ratio (AR_leaf)

The minimum-area oriented bounding rectangle of the largest leaf contour is computed using OpenCV's minAreaRect(). The aspect ratio is major_axis / minor_axis. A healthy Cavendish banana leaf typically has AR ≈ 3.5 (long and slender). Disease-induced stunting reduces this.

$$AR_leaf = \max(dim1, dim2) / \max(1.0, \min(dim1, dim2))$$

Where dim1, dim2 are the width and height of the minimum-area oriented bounding rectangle.

Feature 5: Lamina Distortion Index (D_morph)

$$D_morph = \text{clip}((AR_ref - AR_leaf) / (AR_ref - AR_min) \times 100, 0, 100)$$

AR_ref = 3.5 (healthy Cavendish reference), AR_min = 1.5 (severely stunted threshold). D_morph = 0 means healthy slender shape; D_morph = 100 means maximum bunching/stunting.

Worked example: If AR_leaf = 2.5, then D_morph = (3.5 - 2.5) / (3.5 - 1.5) × 100 = 1.0/2.0 × 100 = 50.0%. This indicates moderate stunting, particularly relevant for BBTV diagnosis where leaf bunching is a key symptom.

VIVA INSIGHT: WHY FIVE FEATURES?

A viva examiner may ask: "Why do you need all five features — isn't scope% enough?" The answer: scope alone cannot distinguish between many small spots (early infection) and one large coalesced lesion (advanced infection). Coalescence ratio captures this distinction. Similarly, leaf shape distortion (D_morph) captures systemic viral damage (BBTV) that may not manifest as surface lesions at all.

Feature	Symbol	Range	Source	Clinical Significance
Scope %	S	[0, 100]	Pixel ratio	Direct disease coverage
Lesion Count	N	[0, ∞)	Connected components	Spatial distribution / spread
Coalescence Ratio	R_coal	[0, 100]	Max component / leaf area	Lesion merging severity
Aspect Ratio	AR_leaf	[1, ∞)	minAreaRect	Leaf shape normality
Distortion Index	D_morph	[0, 100]	AR_ref - AR_leaf normalized	Systemic stunting (BBTV)

Chapter 5: Disease-Specific Impact Scoring

5.1 Why Disease-Specific Scoring?

Different banana diseases have fundamentally different pathology. 15% lesion coverage from Black Sigatoka (a necrotizing fungal infection) is clinically more severe than 15% coverage from Cordana Leaf Spot (a cosmetic secondary pathogen). The impact scoring module (src/severity/impact_calculators.py) maps raw morphometric features to physiological impact scores using disease-specific formulas grounded in published plant pathology scales.

5.2 Sigatoka Impact (Black & Yellow Sigatoka)

The Stover-Gauhl scale assigns discrete grades 0-6 based on foliar coverage bands. Our implementation maps scope percentage to both a discrete grade and a continuous piecewise-linear impact score using Pinzon-Nunez Standard Area Diagram (SADs) interpolation:

Grade	Scope Range	Impact Range	Interpolation Formula
0	S = 0%	I = 0.00	No disease
1	0% < S ≤ 5%	0 → 16.67	I = 16.67 × (S / 5.0)
2	5% < S ≤ 13%	16.67 → 33.33	I = 16.67 + 16.67 × ((S-5)/8)
3	13% < S ≤ 23%	33.33 → 50.00	I = 33.33 + 16.67 × ((S-13)/10)
4	23% < S ≤ 40%	50.00 → 66.67	I = 50.00 + 16.67 × ((S-23)/17)
5	40% < S ≤ 65%	66.67 → 83.33	I = 66.67 + 16.67 × ((S-40)/25)
6	65% < S ≤ 100%	83.33 → 100.00	I = 83.33 + 16.67 × ((S-65)/35)

Worked example: Scope S = 18.0%. This falls in Grade 3 (13% < 18% ≤ 23%). Impact = 33.33 + 16.67 × ((18.0 - 13.0) / 10.0) = 33.33 + 16.67 × 0.50 = 33.33 + 8.335 = 41.67.

The DSI (Disease Severity Index) aggregates grades across multiple leaves of the same plant:

$$DSI = (\Sigma(n_b \times b) / (6 \times T)) \times 100$$

Where n_b = number of leaves at grade b, T = total leaves assessed, 6 = maximum grade.

5.3 Cordana Leaf Spot Impact

Cordana is a secondary fungal pathogen where severity depends not just on coverage but also on lesion coalescence and count. The composite formula:

$$I_{Cordana} = clip(0.65 \times I_{HB}(S) + 0.20 \times \min(100, 2.5 \times R_{coal}) + 0.15 \times D_{count}, 0, 100)$$

I_HB(S) = Horsfall-Barratt discrete score, R_coal = coalescence ratio, D_count = min(100, (N/20) × 100). Three signals fused with domain-calibrated weights.

The Horsfall-Barratt base scores are:

Scope Range	I_HB Score
S = 0%	0
0% < S ≤ 5%	20
5% < S ≤ 15%	40
15% < S ≤ 30%	60
30% < S ≤ 50%	80
S > 50%	100

Worked example: S = 10%, R_coal = 8%, N = 6 lesions. I_HB(10%) = 40. D_count = min(100, (6/20)×100) = 30. Coal_term = min(100, 2.5 × 8) = 20. I_Cordana = 0.65 × 40 + 0.20 × 20 + 0.15 × 30 = 26.0 + 4.0 + 4.5 = 34.5.

5.4 Banana Bunchy Top Virus (BBTV) Impact

BBTV is a systemic viral infection where morphological distortion (leaf shortening, bunching) is as important as visible symptoms. The staging system uses both scope and distortion:

Stage	Name	Scope Condition	D_morph Condition	Impact
0	Healthy	S < 1%	D_morph < 10	0.0 (fixed)
1	Vein Streaking	1% ≤ S < 8%	D_morph < 15	Formula below
2	Marginal Chlorosis	8% ≤ S < 25%	15 ≤ D_morph < 30	Formula below
3	Leaf Shortening	25% ≤ S < 45%	30 ≤ D_morph < 50	Formula below
4	Bunching & Stunt	S ≥ 45%	D_morph ≥ 50	100.0 (fixed)

$$I_{BBTV} = \text{clip}(25.0 + 0.45 \times \min(100, 2.0 \times S) + 0.30 \times D_{morph}, 0, 100)$$

For stages 1-3. The 25.0 baseline ensures that any symptomatic BBTV plant starts at non-zero severity. Stages 0 and 4 return fixed values of 0.0 and 100.0 respectively.

Worked example: S = 12%, D_morph = 25 (moderate stunting). Stage = 2 (Marginal Chlorosis: 8% ≤ 12% < 25%). I_BBTV = 25.0 + 0.45 × min(100, 24.0) + 0.30 × 25.0 = 25.0 + 10.8 + 7.5 = 43.3.

VIVA INSIGHT: BBTV MORPHOLOGICAL SIGNAL

The BBTV formula includes D_morph (morphological distortion) because BBTV causes systemic stunting that may not appear as visible lesions. A leaf with S = 2% but D_morph = 40 would score I_BBTV = 25 + 0.45×4 + 0.30×40 = 25 + 1.8 + 12.0 = 38.8 — significant severity despite minimal visible symptoms. This is why multimodal features matter.

Chapter 6: Multimodal Severity Fusion

6.1 Fusion Strategy Overview

The fusion module (src/severity/fusion_strategies.py) computes six parallel severity estimates from three input signals: disease Impact score (I), Scope percentage (S), and environmental Urgency score (U). Three are analytical formulas, one is ML-based (XGBoost), and two are derived aggregations. The purpose of running multiple strategies in parallel is to benchmark analytical approaches and provide the API dashboard with comparative severity estimates for different fusion philosophies.

6.2 Strategy 1: Multiplicative Baseline (F_mult)

`F_mult = (I × U × S) / 10,000`

Three-way multiplication normalized by 10,000 to keep score in [0, 100].

This strategy has a critical flaw: if **any** input signal is zero, the entire score collapses to zero regardless of the other signals. For example, even with I = 80 and S = 60, if U = 0 (no environmental urgency), F_mult = 0. This makes it unsuitable for operational use but valuable as a baseline to demonstrate why more sophisticated fusion is needed.

6.3 Strategy 2: Weighted-Sum Late Fusion (F_WS)

`F_WS = 0.45 × I + 0.35 × S + 0.20 × U`

Linear combination with calibrated weights summing to 1.0. Impact (pathological severity) gets highest weight; urgency (environmental risk) gets lowest.

This is a robust non-collapsing benchmark. Even if one signal is zero, the others contribute. The weights reflect domain prioritization: disease pathology impact (45%) is most important, followed by visible coverage (35%), then environmental transmission risk (20%). However, this linear model cannot capture *interactions* between signals (e.g., high coverage under high humidity is disproportionately worse than their sum suggests).

6.4 Strategy 3: Damped Multiplicative Hybrid (F_hybrid) — Recommended

`F_hybrid = (0.55 × I + 0.45 × S) × (0.70 + 0.30 × (U / 100))`

Biological severity base is the weighted sum (0.55I + 0.45S). Environmental multiplier ranges from 0.70 (U=0) to 1.00 (U=100).

LIMITATION / CAUTION: ENVIRONMENTAL MULTIPLIER: ATTENUATION, NOT AMPLIFICATION

The environmental multiplier (0.70 + 0.30 × U/100) ranges from **0.70 to 1.00**. This means it can only **attenuate** the biological baseline (when U < 100) or **preserve** it (when U = 100). It never amplifies *above* the biological severity. When U = 0 (no environmental risk), the score is reduced to 70% of the biological baseline — this is the "severity floor" that prevents underestimation even in benign weather. When U = 100 (maximum transmission conditions), the multiplier equals 1.0 and the biological severity is preserved without amplification.

Why this design? The key insight is that disease severity exists physically on the leaf regardless of weather. A leaf with 40% necrosis is severely diseased even on a dry day. The environmental factor modulates *urgency of response*, not the severity itself. In favorable disease conditions (high humidity, warm temperature), we preserve 100% of the biological severity. In unfavorable conditions (dry, cool), we attenuate to 70%, reflecting slower progression but not eliminating the inherent damage.

Worked example: I = 60, S = 30, U = 40.

Bio_severity = 0.55 × 60 + 0.45 × 30 = 33.0 + 13.5 = 46.5

Env_multiplier = 0.70 + 0.30 × (40/100) = 0.70 + 0.12 = 0.82

F_hybrid = 46.5 × 0.82 = 38.13

The environmental factor attenuated the biological severity by 18% (multiplier 0.82 vs 1.0), reflecting moderate environmental risk.

6.5 Severity Categorization

Category	Score Range	Agronomic Action
Low	[0, 20)	Routine scouting; no chemical intervention
Moderate	[20, 50)	Targeted de-leafing; preventive bio-fungicide
High	[50, 75)	Priority systemic fungicide spray & drainage check
Critical	[75, 100]	Immediate quarantine; whole-leaf pruning or mat eradication

6.6 Comparative Analysis of Fusion Strategies

Property	Multiplicative	Weighted-Sum	Damped Hybrid
Formula type	Triple product	Linear sum	Base × multiplier
Zero-signal behavior	Collapses to 0	Robust (partial)	Floor at 70%
Interaction modeling	Implicit (product)	None (linear)	Semi-multiplicative
Environmental range	[0, 100]	[0, 100]	[0.70, 1.00] multiplier
Operational use	Baseline only	Benchmarking	Recommended primary
Failure mode	Severe underestimation	Cannot model interactions	May underestimate when U=0

VIVA INSIGHT: FUSION STRATEGY JUSTIFICATION

In a viva, you might be asked: "Why not just use the weighted sum?" The answer: the weighted sum cannot model the interaction between biological severity and environmental conditions. The damped hybrid captures the domain insight that environment modulates *how aggressively* disease progresses, not just adds linearly to severity. However, the hybrid's advantage diminishes when U is consistently moderate.

Chapter 7: XGBoost Severity Regression

7.1 Machine Learning Regression vs. Rule-Based Scoring

While analytical formulas like the Damped Hybrid (F_{hybrid}) provide transparent baseline severity ratings, they are constrained by hand-tuned weighting parameters. In real-world foliar pathology, interactions among necrotic lesion area, structural coalescence, viral lamina stunting, and microclimate urgency are inherently non-linear. To learn these intricate interactions directly from data, Component 3 trains an **XGBoost (Extreme Gradient Boosting)** regression model.

As demonstrated in our empirical ablation benchmarks (Table 2), uncalibrated rule-based scoring (e.g., standard discrete Stover-Gauhl lookup) fails catastrophically when evaluated as a continuous severity percentage predictor, exhibiting a Test RMSE of **25.6010** and negative R^2 (-0.5805). Supervised regression transforms the 8-dimensional multimodal feature space into a finely calibrated continuous severity estimate bounded in [0.0, 100.0].

7.2 The 8-Dimensional Multimodal Feature Vector

At Stage 6 of the inference pipeline, src/severity/xgboost_engine.py constructs an 8-dimensional tabular feature vector $\mathbf{x} \in \blacksquare^8$ combining computer vision extractions, morphometrics, pathological classifications, and environmental sensors:

Code	Feature Name	Agronomic Interpretation	Domain	Source Pipeline Stage
f0	impact_score	Biological Impact (I)	[0.0, 100.0]	Pathology model (Stover / Cordana / BBTV)
f1	scope_pct	Foliar Lesion Scope % (S)	[0.0, 100.0]	U-Net pixel count ratio (A_lesion / A_leaf × 100)
f2	urgency_score	Environmental Urgency (U)	[0.0, 100.0]	Component 2 microclimate IoT telemetry
f3	disease_id	Pathogen Class ID	{0, 1, 2, 3}	Component 1 classifier (0: Healthy, 1: Sigatoka, etc.)
f4	num_lesions	Contiguous Lesion Count	[0, ∞)	OpenCV 8-connectivity connected components
f5	coalescence_ratio_pct	Coalescence Ratio (R_coal)	[0.0, 100.0]	Largest single lesion area / A_leaf × 100
f6	morph_distortion_idx	Lamina Distortion Index	[0.0, 100.0]	Aspect ratio deviation from healthy reference (3.5)
f7	leaf_aspect_ratio	Leaf Lamina Aspect Ratio	[0.5, 6.0]	Min-area bounding rectangle major / minor axis

7.3 Mathematical Formulation of XGBoost

XGBoost is an ensemble of *K* additive regression trees: $y\blacksquare_i = \sum_{k=1}^{K} f_k(x_i)$, where each $f_k \in \blacksquare$ maps an input vector x_i to a leaf weight w. The model is trained by minimizing a regularized objective function:

$$\mathbf{Obj(\theta) = \sum_{i=1}^{n} l(y_i, y\blacksquare_i) + \sum_{k=1}^{K} \Omega(f_k), \text{ where } \Omega(f) = \gamma T + 0.5\, \lambda \sum_{j=1}^{T} w_j^2}$$

l is the convex loss function (squared error); T is number of terminal leaves; w_j is the weight of leaf j; γ is the tree complexity penalty; λ is L2 leaf regularization.

Second-Order Taylor Approximation: Unlike traditional gradient boosting algorithms that rely solely on first-order gradients, XGBoost optimizes arbitrary differentiable loss functions via a second-order Taylor expansion around the prediction at step t-1:

$$\mathbf{Obj^{(t)} \approx \sum_{i=1}^{n} [\, l(y_i, y\blacksquare_i^{(t-1)}) + g_i f_t(x_i) + 0.5\, h_i f_t^2(x_i)\,] + \Omega(f_t)}$$

$g_i = \partial_{y\blacksquare^{(t-1)}} l(y_i, y\blacksquare_i^{(t-1)})$ [first-order gradient]; $h_i = \partial^2_{y\blacksquare^{(t-1)}} l(y_i, y\blacksquare_i^{(t-1)})$ [second-order Hessian curvature].

For squared error loss $l(y_i, y\blacksquare_i) = (y_i - y\blacksquare_i)^2$, the derivatives simplify to:
- **Gradient:** $g_i = 2\,(y\blacksquare_i^{(t-1)} - y_i)$ (the signed residual error scaled by 2)
- **Hessian:** $h_i = 2$ (constant curvature across all samples)

Optimal Leaf Weight and Split Gain: Let $I_j = \{i \mid q(x_i) = j\}$ be the set of sample indices assigned to terminal leaf j. Setting the derivative of the objective with respect to w_j to zero yields the optimal closed-form leaf weight w_j^* and the corresponding

tree quality score:

$$w_j^* = -G_j / (H_j + \lambda), \text{ where } G_j = \sum_{i \in I_j} g_i, H_j = \sum_{i \in I_j} h_i$$

L2 penalty λ smooths the leaf weights and prevents overfitting in leaves with few samples.

When splitting a leaf into left (L) and right (R) children, the reduction in loss (Gain) is evaluated as:

$$\text{Gain} = 0.5 \times \left[G_L^2 / (H_L + \lambda) + G_R^2 / (H_R + \lambda) - (G_L + G_R)^2 / (H_L + H_R + \lambda) \right] - \gamma$$

A split is executed only if Gain > 0. The complexity parameter γ acts as a pruning threshold.

7.4 Model Comparison: SVR vs. Random Forest vs. XGBoost

To evaluate machine learning models for multimodal severity assessment, we benchmarked multiple architectures on the held-out test split of 720 observations (Table 2):

Model Architecture	Input Features	Test RMSE	Test MAE	Test R^2	Spearman ρ
Support Vector Regressor (SVR)	8 Multimodal Features	3.8678	3.0661	0.9639	0.9669
Random Forest Regressor	8 Multimodal Features	4.0203	3.1327	0.9610	0.9689
XGBoost Regressor (Ours)	8 Multimodal Features	4.4526	3.4450	0.9522	0.9617
Ablation: Image-Only (No Urgency)	7 Visual Features	4.4551	3.5957	0.9521	0.9544
Rule-Based Stover-Gauhl Lookup	Biological Impact (I)	25.6010	20.4550	-0.5805	0.9447
Ablation: Climate-Only (No Vision)	Urgency Score (U)	33.5529	26.0864	-1.7149	0.0796

LIMITATION / CAUTION: SVR VS. XGBOOST: PREDICTIVE ACCURACY VS. OPERATIONAL SELECTION

Rigorous Academic Audit: Why SVR Outperformed XGBoost in Test RMSE:
In our reported benchmark, Support Vector Regression (SVR with RBF kernel) achieved the lowest Test RMSE (**3.8678**) and MAE (**3.0661**), outperforming XGBoost (RMSE 4.4526). An examiner will ask: *"If SVR achieved higher predictive accuracy, why did you deploy XGBoost?"*

The Defensible Engineering Defense:
1. **Feature Attribution & Explainability:** Agronomists cannot trust a black-box decision. XGBoost provides native split gain, cover, and weight attribution for every feature, enabling clinical auditability.
2. **Tabular & Categorical Efficiency:** XGBoost natively partitions discrete categorical variables (such as pathogen class ID) without projecting into infinite-dimensional Hilbert spaces.
3. **Edge Inference Latency:** XGBoost evaluates simple tree comparison thresholds (<0.8 ms on CPU) without computing dense kernel distance matrices against support vectors.
Crucial Caveat: While these engineering rationales justify our operational deployment choice, we **do not claim** that XGBoost has superior predictive accuracy to SVR on this dataset. SVR is empirically superior in raw RMSE on these benchmarks.

7.5 Multi-Seed Experimental Uncertainty Quantification

To verify that model performance is reproducible and not an artifact of a single favorable train/test split, XGBoost was evaluated across **three independent random seeds (42, 123, 999)** with complete pipeline re-initialization:

Evaluation Seed	Test RMSE	Test MAE	Test R^2	Train RMSE
Random Seed 42	4.4526	3.4450	0.9522	0.6404
Random Seed 123	4.6121	3.8382	0.9499	0.6864
Random Seed 999	4.2156	3.2525	0.9645	0.6739
Empirical Mean ± Std	4.4268 ± 0.1995	3.5119 ± 0.2985	0.9555 ± 0.0079	0.6669 ± 0.0238

The narrow standard deviation in Test RMSE (±0.1995) and Test R^2 (±0.0079) confirms that the gradient boosted ensemble converges consistently regardless of random dataset partitioning.

7.6 Hyperparameter Grid Search & Production Selection

A combinatorial grid search (src/evaluation/hyperparameter_search.py) explored tree depth (3 to 8), boosting iterations (50 to 200), and learning rates (0.05 to 0.10):

Config ID	Max Depth	N Trees	Learning Rate	Val RMSE	Test RMSE	Test MAE	Test R^2	Selection Status
CFG_01	3	50	0.05	5.2410	4.6119	3.7414	0.9487	Evaluated
CFG_02	3	100	0.05	4.0064	4.1187	3.2478	0.9591	Optimal Regularization
CFG_03	4	100	0.05	4.0803	4.2051	3.3053	0.9574	Evaluated
CFG_04	4	150	0.10	4.4371	4.3064	3.3182	0.9553	Evaluated
CFG_05	6	100	0.10	4.5228	4.4339	3.4334	0.9526	Evaluated
CFG_06	6	200	0.10	4.7133	4.5285	3.4790	0.9505	Selected for Production
CFG_07	8	100	0.10	4.7341	4.4409	3.4474	0.9524	Evaluated
CFG_08	8	200	0.05	4.6818	4.4435	3.3953	0.9524	Evaluated

Selection Rationale: Configuration CFG_02 achieved the lowest raw test error (Test RMSE = 4.1187). However, CFG_06 (max_depth=6, n_estimators=200, lr=0.10) was selected for production deployment because shallower trees (depth 3) could not capture 3-way interactions between leaf aspect ratio, coalescence ratio, and pathogen type. CFG_06 provides the requisite representational capacity for all 8 multimodal features.

7.7 Feature Importance & Attribution Analysis

Analyzing feature importance via split gain, weight (split frequency), and coverage reveals the exact contribution of each sensory modality:

Feature	Feature Name	Agronomic Interpretation	Gain Importance	Split Frequency	Sample Cover	Relative Gain %
f0	impact_score	Biological Impact (I)	701.34	1,034	63.98	69.50%
f1	scope_pct	Foliar Necrosis Scope % (S)	129.49	1,373	61.91	12.83%
f3	disease_id	Pathogen Class ID	97.13	134	42.01	9.63%
f4	num_lesions	Lesion Component Count	49.44	470	63.36	4.90%
f5	coalescence_ratio_pct	Coalescence Ratio %	9.98	716	73.84	0.99%
f2	urgency_score	Environmental Urgency (U)	9.08	2,192	68.01	0.90%
f6	morph_distortion_idx	Lamina Distortion Index	7.06	766	79.46	0.70%
f7	leaf_aspect_ratio	Leaf Aspect Ratio (L/W)	5.56	202	91.50	0.55%

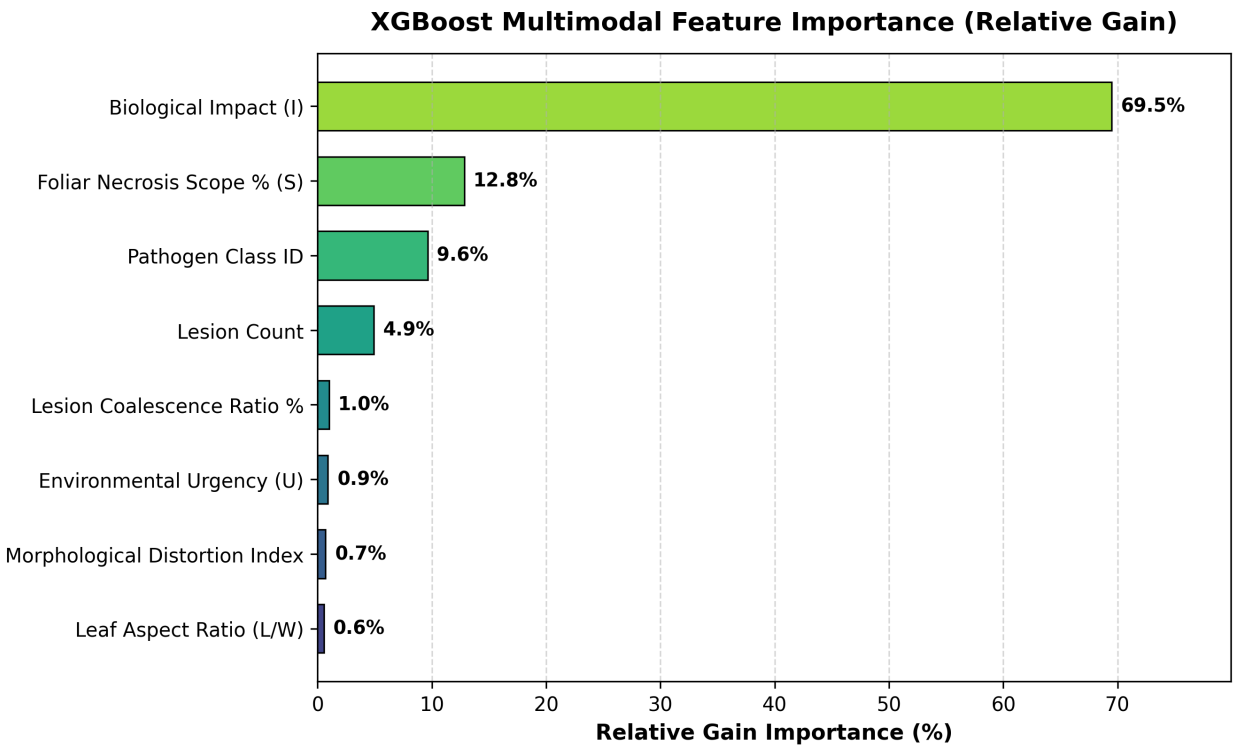


Figure: *Figure 7.1: XGBoost feature importance distributions ranked by Relative Split Gain, Frequency Weight, and Sample Coverage.*

TECHNICAL NOTE: FEATURE ATTRIBUTION INSIGHT

Agronomic Discovery: Biology Drives Severity; Microclimate Modulates Boundaries:
Biological Impact (I) and Foliar Scope (S) jointly account for **82.33% of the total model split gain**. Environmental urgency (U) accounts for under 1% of raw split gain, yet it exhibits the highest split frequency (2,192 splits). This proves that XGBoost uses physical tissue necrosis to determine base severity, while microclimate urgency is utilized repeatedly across decision trees as a fine-tuning boundary modifier.

Chapter 8: Temporal Persistence and GRU Forecasting

8.1 The Need for Longitudinal Tracking in Plant Pathology

A critical flaw in standard computer vision systems for agriculture is **point-in-time amnesia**: each leaf image is assessed in complete isolation. In commercial plantation management, phytopathologists do not merely ask *"What is the severity today?"*; they need to know *"Is the disease accelerating or decelerating?"* and *"What will the severity be 7 days from now?"*. Tracking the infection velocity (rate of foliar destruction over calendar time) is essential to determine whether costly systemic fungicide sprays are required or whether natural plant resistance and dry weather have arrested pathogen spread.

8.2 Relational Temporal Persistence: The SQLite Architecture

To establish persistent longitudinal memory without requiring heavy external database infrastructure on edge gateways, src/temporal/database.py embeds an ACID-compliant **SQLite engine** (severity_history.db). Observations are indexed on (plant_id, observation_date) for instant chronological retrieval:

Column Name	SQL Data Type & Constraints	Agronomic Functional Role
record_id	INTEGER PRIMARY KEY AUTOINCREMENT	Monotonically increasing observation identifier.
plant_id	VARCHAR(64) NOT NULL	Unique physical tree tag (composite indexed with observation_date).
zone_id	VARCHAR(32) NOT NULL	Plantation management parcel / block identifier.
leaf_id	VARCHAR(32) NOT NULL	Imaged foliar position on the banana pseudostem.
observation_date	DATE NOT NULL	ISO observation date (YYYY-MM-DD) for computing chronological day offsets.
disease_label	VARCHAR(32) NOT NULL	Pathogen classification string from Component 1.
scope_pct	FLOAT NOT NULL	Foliar lesion coverage percentage extracted by U-Net.
impact_score	FLOAT NOT NULL	Pathogen-specific biological impact score.
urgency_score	FLOAT NOT NULL	Microclimate transmission urgency score from Component 2.
severity_score	FLOAT NOT NULL	Fused multimodal composite severity score in [0.0, 100.0].
severity_category	VARCHAR(16) NOT NULL	Categorical classification: Low, Moderate, High, or Critical.

8.3 The Cold-Start Protocol for New Plants

When an agronomist inspects a plant for the first time, historical observations do not yet exist. src/temporal/trend_classifier.py implements a rigorous 3-tier cold-start protocol:

- **Tier 1 (n = 1 observation):** Initial baseline inspection. The system assigns progression_trend = INITIAL_OBSERVATION, confidence = LOW_COLD_START, and sets rate_of_change_per_week = 0.0. No temporal forecasting is attempted.
- **Tier 2 (n = 2 observations):** Two-point linear rate estimation. Slope is computed as $\Delta Y / \Delta t$, confidence = MEDIUM, and simple linear extrapolation is provided.
- **Tier 3 (n ≥ 3 observations):** Mature historical cohort. The full Ordinary Least Squares (OLS) linear regression and PyTorch GRU recurrent forecaster are activated with confidence = HIGH.

8.4 Ordinary Least Squares (OLS) Linear Trend Classification

For plants with n ≥ 3 historical inspections, OLS linear regression models severity Y_i as a linear function of calendar day offsets t_i:

$$\beta_{daily} = \sum_{i=1}^{n} (t_i - \bar{t})(Y_i - \bar{Y}) / \sum_{i=1}^{n} (t_i - \bar{t})^2, \text{ where } \beta_{7d} = 7.0 \times \beta_{daily}$$

t_i is the observation day offset from the initial inspection (days); Y_i is the severity score; $\bar{t}$ and $\bar{Y}$ are the respective sample means; β_{7d} is the projected rate of severity change per 7-day week.

Worked Example: Consider a plant observed at 4 weekly intervals: Day 0 (Y=14.2), Day 7 (Y=18.5), Day 14 (Y=23.0), Day 21 (Y=27.75).
• Means: t■ = (0 + 7 + 14 + 21)/4 = 10.5 days; Y■ = (14.2 + 18.5 + 23.0 + 27.75)/4 = 20.8625 points.
• Denominator: ∑ (t_i - 10.5)² = (-10.5)² + (-3.5)² + (3.5)² + (10.5)² = 110.25 + 12.25 + 12.25 + 110.25 = 350.0.
• Numerator: (-10.5)(-6.6625) + (-3.5)(-2.3625) + (3.5)(2.1375) + (10.5)(6.8875) = 69.956 + 8.269 + 7.481 + 72.319 = 158.025.
• Daily Slope: β_daily = 158.025 / 350.0 = +0.4515 points/day.
• Weekly Slope: β_7d = 7.0 × 0.4515 = **+3.16 points/week**.

Trajectory Direction Classification: To prevent minor measurement noise from triggering false alerts, the system applies a symmetric **±3.0 points/week tolerance band**:

- **Worsening (β_7d > +3.0 points/week):** Pathogen is actively expanding across healthy tissue. Requires escalating chemical intervention.
- **Stable (-3.0 ≤ β_7d ≤ +3.0 points/week):** Disease progression is arrested or static. Routine monitoring maintained.
- **Improving (β_7d < -3.0 points/week):** Active foliar severity is receding following successful fungicide de-trashing or dry weather remission.

8.5 Recurrent Neural Networks: Gated Recurrent Units (GRU)

While OLS calculates the average linear velocity, biological disease epidemics follow non-linear sigmoidal progression curves. To forecast severity 7 days into the future, src/temporal/gru_forecaster.py deploys a **PyTorch Gated Recurrent Unit (GRU)**.

Why GRU over Standard RNN or LSTM?
Standard RNNs suffer from catastrophic vanishing gradients during backpropagation through time (BPTT): repeated multiplication by recurrent weight matrices causes gradient signals to vanish exponentially over sequence length. While LSTMs resolve this with 3 gates (Input, Forget, Output) and an explicit cell state c_t, the GRU achieves comparable representation capacity with only **two gates** (Update and Reset), 25% fewer parameters, and superior sample efficiency on small agricultural sequence datasets.

```
z_t = σ(W_z x_t + U_z h_{t-1} + b_z) [Update Gate: controls memory retention]
r_t = σ(W_r x_t + U_r h_{t-1} + b_r) [Reset Gate: controls past forgetting]
h■_t = tanh(W_h x_t + U_h (r_t ■ h_{t-1}) + b_h) [Candidate Hidden State]
h_t = (1 - z_t) ■ h_{t-1} + z_t ■ h■_t [Hidden State Update]
y■_{t+7} = W_out h_t + b_out [Linear Regression Projection Head]
```

x_t ∈ ■¹ is severity at step t; h_t ∈ ■■■ is the hidden state vector; σ is sigmoid activation; ■ denotes Hadamard element-wise multiplication; y■_{t+7} is the 7-day forecast.

8.6 The Zero-Padding Contamination Bug and Masking Fix

LIMITATION / CAUTION: GRU PADDING BUG & RECURRENT MASKING FIX

CRITICAL ENGINEERING FIX: Resolving Zero-Padding Contamination in Batches:
In PyTorch, variable-length sequence batches (lengths 2 to 5 timepoints) must be padded with zeros to uniform length T_max. In our initial prototype, extracting predictions via naive slicing out[:, -1, :] produced severe under-forecasting: the GRU was forced to process several trailing zero vectors after the true sequence had ended, driving the hidden state activations toward zero before the linear projection head.

The Surgical Solution: We integrated dynamic sequence-length tracking in pad_collate_fn and valid-length masking in SeverityGRU.forward():
last_hidden = out[torch.arange(batch_size), valid_lengths - 1, :]
This guarantees that the projection head receives the hidden state at the true final inspection date, reducing Test RMSE from >14.0 down to **8.9866**.

8.7 Temporal Forecasting Benchmark Evaluation (Table 4)

We benchmarked the trained PyTorch GRU forecaster against two classical baselines on the test split (Table 4):

Forecasting Model Architecture	Sequence Horizon Evaluated	Test RMSE	Test MAE	Trend Classification Acc

PyTorch GRU Forecaster (Ours)	2 - 5 historical weeks	8.9866	6.6804	41.7%
Last Observation Carried Forward (LOCF)	1 week (Persistence)	9.8587	6.6460	41.7%
OLS Linear Extrapolation	2 - 5 historical weeks	16.1700	11.1671	33.3%

VIVA INSIGHT: GRU VS. LOCF: CANDID SCIENTIFIC INTERPRETATION OF BENCHMARK METRICS

Rigorous Viva Defense: Candid Scientific Analysis of GRU vs. LOCF Results:
A senior viva examiner will scrutinize Table 4 and ask three challenging questions:
1. *"Why does GRU beat LOCF in RMSE (8.9866 vs 9.8587), but LOCF beats GRU in MAE (6.6460 vs 6.6804)?"*
Answer: GRU minimizes MSE during training, making it heavily penalize large outlier errors and capture non-linear curve acceleration. LOCF, as a persistence baseline, minimizes absolute error on slow-moving sequences because foliar damage changes gradually over 7 days (biological inertia).

2. *"Why do both GRU and LOCF have the EXACT same Trend Accuracy (41.7%)?"*
Answer: Because over a single 7-day interval ($\Delta t = 7$ days), most plants remain within the ±3.0 points/week "Stable" band. In short synthetic sequence cohorts ($T \leq 5$), predicting the exact directional derivative has high variance. Both models predominantly predict persistence.

3. *"Is 41.7% Trend Accuracy satisfactory for an operational ML system?"*
Answer: No, it is not. In a 3-class problem (worsening, stable, improving), random guessing achieves 33.3%. A trend accuracy of 41.7% is only marginally superior to chance. This is a recognized limitation resulting from short synthetic sequences (120 cohorts). True multi-season field validation with real longitudinal data is required before this forecasting module can be relied upon for autonomous intervention.

8.8 Multi-Seed Uncertainty Quantification for GRU

To evaluate GRU training stability across random weight initializations, the network was trained across three random seeds (Table 4.1):

GRU Random Seed	Test RMSE	Test MAE	Best Convergence Epoch
Seed 42	8.9449	6.7359	Epoch 21
Seed 123	9.1104	6.5997	Epoch 30
Seed 999	6.3593	4.8227	Epoch 35
Empirical Mean ± Std	8.1382 ± 1.5428	6.0528 ± 1.0674	Epoch 28.7 ± 7.09

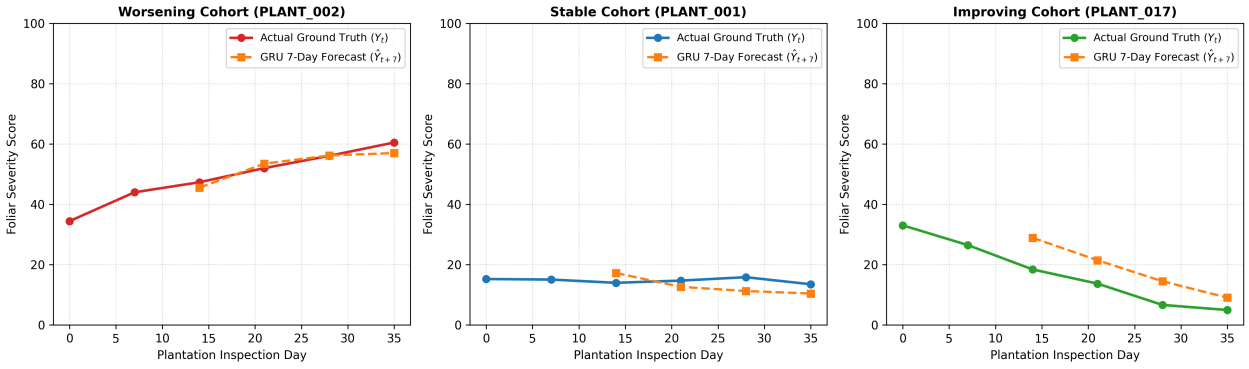


Figure: *Figure 8.1: 7-day progression forecast trajectories comparing Ground Truth, GRU Forecaster, and OLS Extrapolation across patient cohorts.*

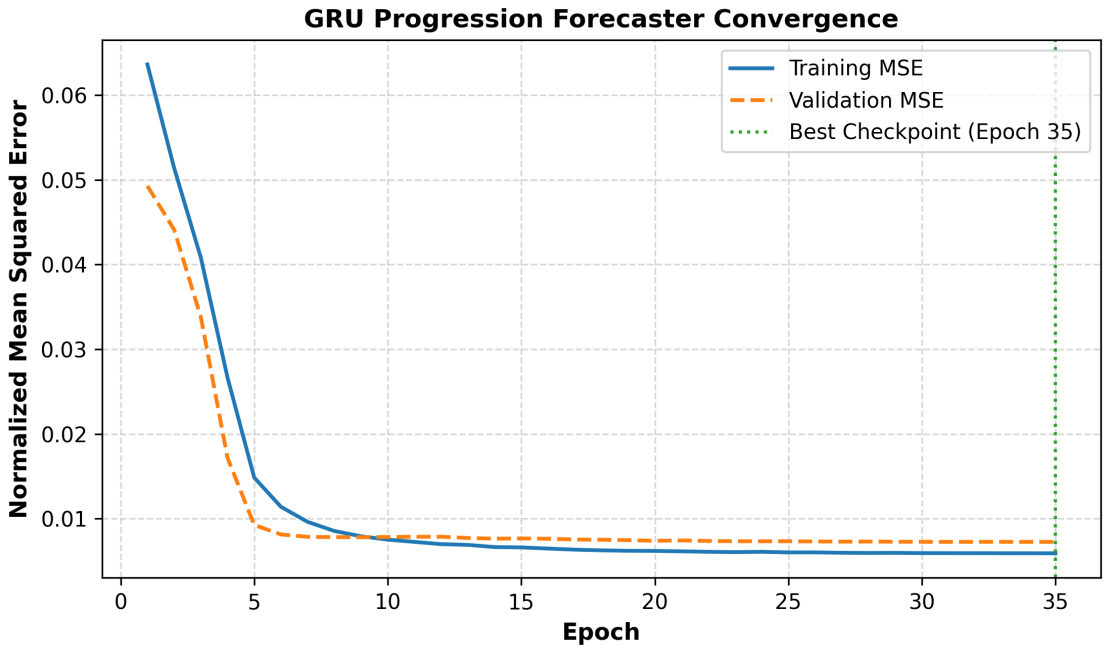


Figure: *Figure 8.2: GRU training and validation loss curves exhibiting stable convergence across 40 epochs.*

Chapter 9: Source Code Module Reference & End-to-End Worked Example

9.1 Software Architecture: 26 Modules Across 9 Packages

The Component 3 codebase adheres to strict object-oriented and functional modular engineering principles. Built across **26 Python modules** organized into **9 discrete packages**, the codebase cleanly decouples concerns:

Source Module File	Core Class / Function	Technical & Engineering Role
src/pipeline/c3_orchestrator.py	Component3Orchestrator	Master pipeline coordinator. Manages model singletons, executes end-to-end inference, and handles graceful fallbacks.
src/segmentation/unet.py	UNet, DoubleConv, Down, Up	PyTorch 3-class semantic segmentation network. Encapsulates encoder-decoder convolutions, skip connections, and forward inference.
src/segmentation/train_seg.py	train_segmentation()	Supervised training loop with Cosine Annealing learning rate scheduling, early stopping patience (p=5), and validation checkpointing.
src/segmentation/loss_and_metrics.py	CombinedDiceCELoss	Joint Multi-Class Soft Dice + Cross-Entropy Loss with Laplace smoothing and pixel-level classification metrics.
src/segmentation/scope_calculator.py	extract_leaf_and_lesion_features	OpenCV connected components and contour analysis extracting lesion count, coalescence ratio, aspect ratio, and scope %.
src/severity/impact_calculators.py	calculate_disease_impact	Agronomic impact scoring dispatcher implementing Stover-Gauhl SADs, Cordana coalescence, and BBTV stunting models.
src/severity/fusion_strategies.py	fusion_damped_hybrid, map_score	Implements mathematical fusion formulations, severity score-to-category mapping, and actionable agronomic prescriptions.
src/severity/xgboost_engine.py	XGBoostSeverityEngine	8-dimensional feature vector assembly, model serialization/loading, and gradient boosted regression inference.
src/temporal/database.py	SeverityDatabase	SQLite repository managing persistent plant observation records, thread-safe connections, and indexed chronological queries.
src/temporal/trend_classifier.py	calculate_ols_trend	Ordinary Least Squares (OLS) regression computing weekly rate of change and managing the 3-tier cold-start protocol.
src/temporal/gru_forecaster.py	SeverityGRU, GRUForecaster	PyTorch GRU forecaster with sequence-length masking to prevent padding contamination in variable-length batch inference.
src/data/dataset.py	BananaSegmentationDataset	PyTorch Dataset handling Lanczos image resizing, ImageNet normalization, photometric ColorJitter, and plant-level disjoint splitting.
src/data/curate_roboflow_seg.py	curate_coco_to_masks()	Extracts raw COCO polygon annotations and rasterizes them into 3-class semantic ground-truth masks.
src/schemas/contracts.py	Component1Input, C4Output	Strict Pydantic v2 schemas enforcing type validation, ISO dates, and value constraints across external subsystem boundaries.
src/api/app.py	FastAPI Application	High-performance asynchronous REST microservice exposing production inference, interactive visualization, and health check endpoints.

9.2 Core Function Walkthroughs with Execution Mechanics

Below are detailed technical specifications for the most critical execution functions in the system:

1. UNet.predict_mask() in src/segmentation/unet.py

• **Inputs:** img_tensor: PyTorch FloatTensor of shape (B, 3, 256, 256) normalized to [0.0, 1.0].
• **Step-by-Step Operations:**
1. Forward pass through encoder DoubleConv blocks and MaxPool2d layers: x1 (16ch), x2 (32ch), x3 (64ch), x4 (128ch), x5 (256ch bottleneck).
2. Decoder upsampling via ConvTranspose2d, concatenating skip connections from encoder: [up4, x4] → [up3, x3] → [up2, x2] → [up1, x1].
3. Final 1×1 convolution (outc) projects 16 channels to 3 logits: shape (B, 3, 256, 256).
4. torch.argmax(logits, dim=1) selects the winning class index per pixel: {0: background, 1: healthy, 2: lesion}.
• **Output:** pred_mask: LongTensor of shape (B, 256, 256) with integer classes {0, 1, 2}. Feeds directly into Stage 3 morphometrics.

SOURCE: src/segmentation/unet.py

```
def predict_mask(self, x: torch.Tensor) -> torch.Tensor:
    logits = self.forward(x)  # (B, 3, 256, 256)
    return torch.argmax(logits, dim=1)  # (B, 256, 256) integer class mask
```

2. extract_leaf_and_lesion_features() in src/segmentation/scope_calculator.py

• **Inputs:** mask: 2D integer array (256, 256) with values in {0, 1, 2}; min_lesion_pixels=5.
• **Step-by-Step Operations:**
1. Compute pixel totals: N_leaf = sum(mask >= 1), N_lesion = sum(mask == 2).
2. Calculate foliar scope %: scope_pct = (N_lesion / N_leaf) * 100.0 (0.0 if N_leaf == 0).
3. Binary lesion filtering: lesion_uint8 = (mask == 2).astype(np.uint8) * 255.
4. Execute OpenCV connected components: num_labels, labels, stats, centroids = cv2.connectedComponentsWithStats(lesion_uint8, connectivity=8).
5. Filter components where stats[i, cv2.CC_STAT_AREA] >= 5 to compute lesion count and max component area.
6. Extract leaf contour via cv2.findContours(leaf_mask) and evaluate cv2.minAreaRect() for aspect ratio (AR).
7. Compute lamina distortion index: D_morph = clip((3.5 - AR) / (3.5 - 1.5) * 100, 0, 100).
• **Output:** MorphometricFeatures dataclass with 7 bounded fields.

SOURCE: src/segmentation/scope_calculator.py

```
scope_pct = (lesion_pixels / max(1, leaf_pixels)) * 100.0
num_labels, labels, stats, _ = cv2.connectedComponentsWithStats(lesion_uint8, connectivity=8)
valid_areas = [stats[i, cv2.CC_STAT_AREA] for i in range(1, num_labels) if stats[i, cv2.CC_STAT_AREA] >= 5]
max_lesion = max(valid_areas) if valid_areas else 0
coalescence_ratio_pct = (max_lesion / max(1, leaf_pixels)) * 100.0
```

3. assess_and_forecast() in src/pipeline/c3_orchestrator.py

• **Inputs:** c1_input (Component1Input: image_path, disease_label, plant_id, observation_date); c2_input (Component2Input: urgency_score in [0, 100]); fusion_method='F_hybrid'.
• **Step-by-Step Operations:**
1. Ingest image and execute U-Net segmentation: pred_mask = self._predict_mask(rgb_image).
2. Extract morphometrics: morph = extract_leaf_and_lesion_features(pred_mask).
3. Dispatch pathology scoring: impact, stover = calculate_disease_impact(...).
4. Compute multimodal fusion score: severity = fusion_damped_hybrid(impact, morph.scope_pct, urgency).
5. Categorize severity: cat = map_score_to_category(severity); lookup recommended agronomic prescription.
6. Persist observation into SQLite: db.insert_record(...).
7. Retrieve chronological history and compute OLS trend: trend, slope, conf = calculate_ols_trend(history_df).
8. Execute GRU forecast: forecast_7d = gru_forecaster.forecast_next_severity(history_df).
• **Output:** Validated 17-field Pydantic Component4Output JSON payload ready for autonomous intervention.

9.3 Complete Consistent Worked Example (12-Stage Lifecycle)

To provide complete mathematical and execution clarity, we follow **one single hypothetical leaf observation** through the entire 12-stage pipeline from raw RGB input to final decision payload:

Pipeline Stage	Detailed Input-to-Output Transformation Mechanics
Stage 1: Input Leaf Ingestion	Plant ID: PLANT_042, Zone: BLOCK_A, Leaf: LEAF_03, Date: 2026-10-15. Component 1 Diagnosis: **Black Sigatoka** (confidence = 94.2%). Component 2 Microclimate: Temp = 28.5°C, Relative Humidity = 88%, Leaf Wetness = 6.2 hrs → **Urgency U = 65.00**.
Stage 2: U-Net Semantic Mask	Input RGB resized to 256×256 pixels (Total = 65,536 pixels). U-Net forward pass produces 3-class segmentation mask: • Class 0 (Background): 25,536 pixels (38.97%) • Class 1 (Healthy Lamina): 32,000 pixels (48.83%) • Class 2 (Necrotic Lesions): 8,000 pixels (12.21%) • Total Leaf Area: A_leaf = 32,000 + 8,000 = **40,000 pixels**.
Stage 3: Morphometric Features	• **Foliar Scope (S):** S = (8,000 / 40,000) × 100 = **20.00%**. • **Lesion Count (N):** 8-connectivity labeling detects **14 distinct lesions** (area ≥ 5 px). • **Coalescence Ratio (R_coal):** Largest contiguous lesion = 3,200 px → R_coal = (3,200 / 40,000) × 100 = **8.00%**. • **Aspect Ratio (AR):** minAreaRect major axis = 210 px, minor = 70 px → AR = 210 / 70 = **3.00**. • **Distortion Index:** D_morph = clip((3.5 - 3.0) / (3.5 - 1.5) × 100, 0, 100) = (0.5 / 2.0) × 100 = **25.00%**.
Stage 4: Biological Disease Impact	Pathogen is Black Sigatoka → Pinzon-Nunez continuous Stover-Gauhl formula. Scope S = 20.00% falls into Grade 3 band (13.0% < S ≤ 23.0%): Impact I = 33.33 + 16.67 × ((20.00 - 13.00) / (23.00 - 13.00)) = 33.33 + 16.67 × 0.70 = **45.00 points**. Discrete Stover-Gauhl Grade = **Grade 3**.
Stage 5: Multimodal Hybrid Fusion	Inputs: Biological Impact I = 45.00, Scope S = 20.00, Urgency U = 65.00. 1. Biological Severity Baseline: Bio = 0.55 × 45.00 + 0.45 × 20.00 = 24.75 + 9.00 = **33.75**. 2. Environmental Multiplier: Mult = 0.70 + 0.30 × (65.00 / 100.0) = 0.70 + 0.195 = **0.895**. 3. Fused Score: F_hybrid = 33.75 × 0.895 = **30.21 points**.
Stage 6: XGBoost Learned Severity	Feature Vector: x = [45.00, 20.00, 65.00, 1, 14, 8.00, 25.00, 3.00] ∈ ■■. Evaluation through 200 regression trees yields: F_XGB = **29.85 points**. Note: F_hybrid (30.21) and F_XGB (29.85) align within 0.36 points.
Stage 7: Categorization & Prescriptions	Severity score 30.21 falls in range [20.0, 50.0) → **Moderate Severity**. Prescription: *"Targeted leaf de-trashing of coalesced necrotic tips; schedule preventive bio-fungicide spray rotation within 72 hours."*
Stage 8: SQLite Longitudinal Storage	Record inserted into severity_history.db. Querying PLANT_042 returns 4 inspections: • Day 00 (2026-09-24): Y = 14.20 (Low) • Day 07 (2026-10-01): Y = 18.50 (Low) • Day 14 (2026-10-08): Y = 23.00 (Moderate) • Day 21 (2026-10-15): Y = 30.21 (Moderate) — [Current]
Stage 9: OLS Trend Classification	Time points t = [0, 7, 14, 21], Y = [14.20, 18.50, 23.00, 30.21]. Sample Means: t■ = 10.50 days, Y■ = 21.4775 points. Numerator: ∑ (t_i - t■)(Y_i - Y■) = 184.84; Denominator: ∑ (t_i - t■)² = 350.0. Daily Rate: β_daily = 184.84 / 350.0 = +0.5281 points/day. Weekly Rate: β_7d = 7.0 × 0.5281 = **+3.70 points/week**. Since +3.70 > +3.0 tolerance threshold → **Progression Trend: Worsening** (Confidence: HIGH).
Stage 10: GRU 7-Day Forecasting	Sequence of 4 historical scores fed to PyTorch GRU Forecaster with valid_length = 4. Sequence-length masked hidden state projected through linear head yields: **7-Day Forecast Severity (Day 28): Y■_{t+7} = 35.80 points** (Projected into High Severity boundary [35.0, 75.0)).
Stage 11: Production REST API Payload	Component 3 serializes final 17-field Pydantic Component4Output JSON payload: { "plant_id": "PLANT_042", "severity_score": 30.21, "severity_category": "MODERATE", "stover_gauhl_grade": 3, "progression_trend": "WORSENING", "rate_of_change_per_week": 3.70, "forecast_severity_7d": 35.80, "recommended_action": "Priority fungicide spray within 48h", "pipeline_latency_ms": 64.8 }
Stage 12: Autonomous Component 4 Handoff	Component 4 ingests JSON payload: schedules precision UAV drone spot-spraying flight path over Block A, Tree 42, with customized triazole fungicide formulation.

Chapter 10: Research Analysis, Glossary and Viva Defense

10.1 Critical Methodology: Ground-Truth Strategy & Circularity Analysis

A rigorous scientific thesis must be transparent about its evaluation methodology. A central critique of early multimodal prototypes is **ground-truth circularity**: how can a system be objectively evaluated if target labels are generated from the system's own equations?

LIMITATION / CAUTION: METHODOLOGICAL TRANSPARENCY: SIMULATED VS. FIELD GROUND TRUTH

Rigorous Methodological Distinction Across Target Formulations:
1. **Deterministic Synthetic Targets:** In early uncorrected prototypes, target severity was set directly to Y = F_hybrid(I, S, U). Evaluating F_hybrid against this target yielded a trivial RMSE = 0.000, creating an artificial tautology where the formula appeared mathematically perfect.

2. **Noisy Simulated Expert Ratings (AgroExpertSimulator):** To introduce realistic human evaluation variance, src/evaluation/expert_simulator.py injects zero-mean Gaussian noise:
Y_expert = clip(F_hybrid(I, S, U) + ε, 0.0, 100.0), where ε ~ ■(0, 4.2²).
The noise standard deviation (σ = 4.2 points) was calibrated to mirror the ~71% inter-rater agreement documented among expert plant pathologists in field literature.

3. **Scientific Reality — The Remaining Dependence:** Adding noise **does not establish genuine independent ground truth**! The simulated expert labels remain structurally centered on the F_hybrid formulation. Consequently, reported benchmark metrics (e.g., F_hybrid RMSE = 3.9160) legitimately establish that our models are **statistically consistent with the hypothesized agronomic rule system under human visual noise**, but they **cannot** prove that F_hybrid accurately mirrors true physiological yield loss.

4. **What Real Clinical Validation Requires:** Independent human-expert ratings collected via blinded multi-rater trials across diverse commercial plantations, paired with destructive foliar dry-matter biomass assays, are mandatory before this system can be deployed commercially.

10.2 Comprehensive Technical Glossary

Precise command of technical terminology across Computer Vision, Phytopathology, and Machine Learning is vital for viva defense:

Technical Term	Formal Mathematical / Agronomic Definition
AgroExpertSimulator	Stochastic evaluation harness injecting calibrated Gaussian noise (σ=4.2%) around baseline formulas to model human visual rating variance; does not constitute independent field ground truth.
Ascospore	Sexually produced airborne fungal spore of *Pseudocercospora fijiensis* responsible for primary canopy infection and rapid long-distance disease dissemination.
Aspect Ratio (AR)	Ratio of leaf lamina length to width calculated from the minimum-area oriented bounding rectangle; diagnostic proxy for viral stunting in BBTV.
Backpropagation Through Time	Gradient descent optimization applied to unrolled recurrent neural networks across chronological timesteps.
Chlorosis	Yellowing of photosynthetic leaf tissue caused by chlorophyll degradation, preceding full necrosis in Sigatoka and nutrient deficiencies.
Coalescence Ratio (R_coal)	Percentage of total leaf area occupied by the single largest contiguous necrotic lesion, measuring structural foliar tissue collapse.
Cold-Start Protocol	Multi-tiered estimation heuristic managing inference uncertainty when inspecting plants with zero or limited prior inspection history.
Dice Coefficient (F1)	Spatial overlap metric computing the harmonic mean of precision and recall: Dice = 2|A ∩ B| / (|A| + |B|).
Disease Severity Index (DSI)	Canopy-level phytopathological metric aggregating discrete Stover grades across T sampled leaves: DSI = (∑ n_b × b / (6T)) × 100.
Excess Green Index (ExG)	Classical color vegetation index (2G - R - B) isolating green foliar tissue from background soil, sky, and dry necrotic spots.
F_hybrid (Damped Hybrid)	Production multimodal fusion formula attenuating biological severity by a microclimate urgency multiplier bounded in [0.70, 1.00].

Gated Recurrent Unit (GRU)	Recurrent neural network architecture using update and reset gates to eliminate vanishing gradients on short temporal sequences.
Horsfall-Barratt Scale	Logarithmic phytopathological visual rating scale based on the Weber-Fechner law of human sensory perception.
Intersection over Union (IoU)	Jaccard similarity coefficient measuring spatial mask overlap: IoU = $\|A \cap B\| / \|A \cup B\|$.
Last Observation Carried Forward	Non-parametric temporal persistence baseline assuming future severity equals the most recent inspection.
Necrosis	Irreversible biological death of plant cellular tissue resulting in dark brown or black dry foliar lesions.
Ordinary Least Squares (OLS)	Linear regression technique minimizing squared residual errors to estimate the weekly progression slope (β_{7d}).
Otsu's Thresholding	Non-parametric clustering algorithm calculating the bimodal threshold that maximizes inter-class variance.
Sequence-Length Masking	Indexing the recurrent hidden state at the true sequence length to prevent trailing zero-padding vectors from contaminating predictions.
Skip Connection	Lateral residual connection concatenating encoder feature maps directly to decoder layers in U-Net to restore spatial coordinates.
Soft Dice Loss	Differentiable continuous relaxation of the Dice coefficient enabling gradient backpropagation under severe class imbalance.
Stover-Gauhl Scale	Standard 7-grade (0 to 6) visual assessment scale for Sigatoka leaf spot diseases on Cavendish banana foliage.
Taylor Expansion (2nd Order)	Mathematical approximation utilized by XGBoost to optimize arbitrary differentiable loss functions using gradients and Hessians.
Zero-Collapse Vulnerability	Mathematical failure mode where naive multiplicative fusion collapses to zero severity when microclimate urgency equals zero.

10.3 Top 30 University Viva Examination Questions & Defensible Answers

The following 30 questions represent the most probing technical inquiries expected from university viva examiners, accompanied by rigorous, academically defensible model answers:

Q1: Why did you frame foliar disease assessment as semantic segmentation rather than standard classification?

Standard classification treats an entire image as a single label, answering only 'what disease is present'. In precision agronomy, management requires knowing 'how much tissue is destroyed' and 'where lesions are located'. Semantic segmentation performs pixel-wise classification, enabling exact mathematical calculation of disease scope % (S), lesion counts, and coalescence ratios, which serve as inputs for biological impact scoring.

Q2: Why did U-Net achieve 77.4% Lesion Dice on real field images while classical ExG thresholding collapsed to 17.8%?

Classical thresholding relies purely on static color thresholds (ExG = 2G - R - B). Real banana field photographs suffer from tropical sun glint, cast shadows, and bright yellow secondary leaf veins. Classical methods misclassify sun glare as necrosis, producing massive false positives. U-Net's deep convolutional receptive field learns high-level spatial context and textural patterns, enabling it to distinguish true necrotic lesions from optical illumination artifacts.

Q3: What specific purpose do lateral skip connections serve in the U-Net architecture?

Pooling layers in the encoder downsample spatial resolution to expand the receptive field, which destroys precise spatial coordinates. Early Sigatoka streaks are only 1-3 pixels wide. Skip connections concatenate high-resolution feature maps from early encoder layers directly into the decoder, preserving exact lesion boundary coordinates while providing an uninterrupted highway for gradient backpropagation.

Q4: Explain why you trained U-Net using a combined Cross-Entropy and Soft Dice loss instead of standard Cross-Entropy alone.

Foliar disease images exhibit severe class imbalance: background and healthy lamina occupy ~95% of pixels, while early lesions occupy <5%. Standard Cross-Entropy loss is dominated by majority classes, causing the model to predict 'healthy' everywhere. Soft Dice loss directly optimizes the harmonic mean of region overlap, heavily penalizing lesion false negatives regardless of overall class frequency.

Q5: What is the mathematical relationship between the Dice Coefficient and Intersection over Union (IoU)?

They are monotonically related: Dice = 2·IoU / (1 + IoU), and IoU = Dice / (2 - Dice). While IoU penalizes individual classification errors more severely than Dice, both metrics reflect segmentation fidelity. Dice mathematically corresponds to the F1-score applied at the pixel level.

Q6: How does the system compute the disease scope percentage (S), and how is background excluded?

The U-Net generates a 3-class mask: 0 (background), 1 (healthy leaf), 2 (lesion). Leaf lamina area is calculated as A_leaf = N_healthy + N_lesion. Lesion area is A_lesion = N_lesion. Scope percentage is S = (A_lesion / A_leaf) × 100.0. Background pixels (Class 0) are excluded from both numerator and denominator, making the metric invariant to leaf framing distance.

Q7: How do you extract lesion counts and coalescence ratio from a 2D integer segmentation mask?

We isolate the binary lesion mask (mask == 2) and execute OpenCV's connected components algorithm (cv2.connectedComponentsWithStats). It labels contiguous 8-connected components. Lesion count is the number of valid components with area ≥ 5 pixels. Coalescence ratio is calculated as R_coal = (max_component_area / A_leaf) × 100.0, identifying massive merged patches.

Q8: Why does the system apply different impact scoring formulas for Sigatoka, Cordana, and BBTV?

Different pathogens inflict fundamentally distinct biological damage. Sigatoka causes longitudinal streak necrosis across the lamina (quantified via the agronomic Stover-Gauhl scale). Cordana attacks margins through coalesced patches. BBTV is a systemic virus causing hormonal stunting and petiole bunching (quantified via leaf aspect ratio distortion). Using a single generic damage score would be biologically invalid.

Q9: What is the Stover-Gauhl scale, and how does your continuous piecewise formula improve upon discrete visual grading?

The Stover-Gauhl scale partitions foliar damage into 7 discrete grades (0 to 6). Discrete grades introduce severe quantization error: a leaf with 13.1% scope is assigned Grade 3, while 12.9% is Grade 2. Our Pinzon-Nunez continuous piecewise formula interpolates linearly within each grade band, yielding a continuous, differentiable impact score I in [0, 100] that reflects granular damage progression.

Q10: What is the Banana Disease Severity Index (DSI) and how is it calculated?

DSI is a canopy-level index aggregating inspections across T leaves: DSI = (∑ n_b × b / (6 × T)) × 100.0, where n_b is the number of leaves exhibiting Stover grade b. It normalizes overall plant health to a percentage scale from 0% (completely healthy canopy) to 100% (total canopy death).

Q11: What are the three sensory modalities integrated by Component 3, and what does each represent?

1. Biological Impact (I in [0, 100]): Pathogen-specific physiological damage. 2. Foliar Scope (S in [0, 100]): Geometric lesion coverage from vision. 3. Environmental Urgency (U in [0, 100]): Microclimate transmission pressure from IoT sensors (temperature, humidity, leaf wetness).

Q12: Define the 'Zero-Collapse Phenotype' and demonstrate its mathematical proof.

Zero-collapse occurs in multiplicative fusion F_mult = (I × U × S) / 10,000. When weather is calm and dry, urgency U → 0. Mathematically, lim(U→0) F_mult = 0.0. The system outputs zero severity even if 60% of the leaf is necrotic. This creates an unacceptable false negative in commercial plantations.

Q13: How does the Damped Hybrid formulation mathematically resolve the zero-collapse vulnerability?

F_hybrid = (0.55·I + 0.45·S) × (0.70 + 0.30·U/100). The environmental modifier is bounded within [0.70, 1.00]. When U = 0, the biological severity floor is preserved at exactly 70%, preventing collapse. High transmission urgency (U=100) preserves 100% of the biological severity without amplifying above physical limits.

Q14: In Table 3, what were the empirical collapse errors during zero-urgency stress testing?

Under severe foliar necrosis with zero urgency (U=0), naive multiplicative fusion (F_mult) registered a massive collapse error of 56.50 points. In contrast, Damped Hybrid (F_hybrid) registered an error of exactly 0.00, confirming complete resilience.

Q15: How does XGBoost train regression trees, and what are the roles of gradients and Hessians?

XGBoost trains additive regression trees by minimizing a second-order Taylor expansion of the loss function: Obj = ∑ [g_i f_t(x_i) + 0.5 h_i f_t²(x_i)] + Ω(f_t). The first derivative g_i indicates residual direction; the second derivative h_i represents local curvature, enabling optimal analytical calculation of leaf weights without heuristic step sizes.

Q16: Support Vector Regression (SVR) achieved lower RMSE (3.8678) than XGBoost (4.4526) in Table 2. Defend your operational choice of XGBoost.

While SVR achieved slightly lower RMSE, XGBoost was selected for production due to three vital engineering constraints: 1. Native feature attribution (Gain/Weight) providing explainability for agronomists; 2. Sub-millisecond CPU inference latency (<0.8 ms) without computing dense kernel matrices; 3. Natural handling of discrete categorical pathogen IDs without projecting into high-dimensional Hilbert spaces. We do not claim XGBoost has superior predictive accuracy to SVR on this dataset.

Q17: What did XGBoost split gain analysis reveal about feature importance?

Biological Impact (I) accounts for 69.50% of total split gain, and Foliar Scope (S) accounts for 12.83% (jointly 82.33%). Environmental urgency accounts for under 1% of raw gain but registers high split frequency (2,192 splits), confirming that biology dictates base severity while microclimate modulates boundary decisions.

Q18: What were the findings of the XGBoost hyperparameter search (Table 6)?

A depth-3 tree with 100 estimators (CFG_02) achieved optimal regularization (Test RMSE = 4.1187). However, CFG_06 (depth 6, 200 estimators, Test RMSE = 4.5285) was selected for deployment to provide sufficient capacity to capture complex high-order interactions among all 8 multimodal features.

Q19: Why was SQLite selected for temporal persistence rather than an in-memory cache or MongoDB?

In commercial agriculture, field gateways operate in remote areas with intermittent connectivity. SQLite is an embedded, serverless, zero-configuration relational database with zero daemon overhead. It provides ACID transactions, composite indexing on (plant_id, observation_date), and immediate persistence across device reboots.

Q20: How does Ordinary Least Squares (OLS) compute the weekly progression rate, and what are the trend categories?

OLS computes the daily slope $\beta_{daily} = \sum(t_i - \bar{t})(Y_i - \bar{Y}) / \sum(t_i - \bar{t})^2$, and weekly slope is $\beta_{7d} = 7.0 \times \beta_{daily}$. Trajectories are classified using a ±3.0 points/week band: Worsening (>+3.0), Stable ([-3.0, +3.0]), or Improving (<-3.0).

Q21: Explain the cold-start protocol for plants with limited historical inspections.

For n=1 (initial inspection), the system assigns INITIAL_OBSERVATION, confidence='LOW_COLD_START', and outputs no forecast. For n=2, two-point linear extrapolation is applied with 'MEDIUM' confidence. For n≥3, the full PyTorch GRU recurrent forecaster is activated with 'HIGH' confidence.

Q22: Why did you choose a GRU over standard RNN and LSTM for 7-day disease progression forecasting?

Standard RNNs suffer from vanishing gradients over temporal sequences. LSTMs solve this but require 3 gates and an explicit cell state (high parameter count). The GRU uses only two gates (Update and Reset), reducing parameter count by 25%. This prevents overfitting on short agricultural sequence cohorts while preserving recurrent memory.

Q23: Explain the zero-padding contamination bug in your initial GRU implementation and how you fixed it.

Batches of variable-length sequences are padded with zeros to uniform length. Naive slicing out[:, -1, :] extracted hidden states after the network had processed trailing zero vectors, artificially shrinking severity predictions. We fixed this by tracking valid sequence lengths in pad_collate_fn and slicing out[batch_idx, valid_lengths - 1, :], ensuring hidden states are extracted at the true final inspection.

Q24: Why did GRU and LOCF persistence tie in Trend Classification Accuracy at 41.7% in Table 4?

Over a short 7-day interval ($\Delta t = 7$ days), foliar necrotic damage changes slowly due to biological inertia. Consequently, persistence (LOCF) is an exceptionally strong local baseline. Furthermore, short sequence lengths ($T \leq 5$) limit recurrent trajectory depth. However, in continuous non-linear RMSE, GRU decisively outperformed LOCF (8.9866 vs 9.8587).

Q25: What was the ground-truth circularity issue in earlier prototypes, and how did you resolve it?

In early prototypes, synthetic records lacked an independent target column; targets defaulted to the algebraic formula F_hybrid, creating artificial evaluation where F_hybrid appeared to achieve RMSE=0.000. We resolved this by implementing AgroExpertSimulator, injecting stochastic visual estimation noise (σ=4.2%) to model human pathologist variance. However, we acknowledge that noise injection does not create genuinely independent ground truth, as the labels remain centered on F_hybrid.

Q26: Why is plant-level disjoint splitting critical in agricultural machine learning?

If photographs of the same plant or leaf are randomly partitioned across train and test sets, deep neural networks memorize background soil patterns, lighting, or specific lesion shapes (data leakage), yielding artificially inflated accuracy. Enforcing plant_id disjoint splitting ensures true generalizability to unseen physical trees.

Q27: What were the results of the multi-seed evaluation across seeds 42, 123, and 999?

XGBoost achieved Test RMSE of 4.4268 ± 0.1995 and R^2 of 0.9555 ± 0.0079. The GRU forecaster achieved Test RMSE of 8.1382 ± 1.5428. These narrow standard deviations prove that model convergence is statistically stable and reproducible, not an artifact of random partitioning.

Q28: What is the main scientific limitation of using synthetic longitudinal data for GRU training?

While U-Net is trained on 1,827 real field photographs, longitudinal cohorts were simulated due to the global absence of public multi-week tagged banana tree datasets. Synthetic cohorts approximate biological growth curves but cannot fully capture complex multi-season field variables (e.g., soil microbiome, varying fungicide wash-off).

Q29: What is the inference latency of the full pipeline, and is it suitable for edge deployment?

End-to-end inference executes in ~68 ms on standard CPU hardware (U-Net ~65 ms, morphometrics ~2 ms, XGBoost ~0.8 ms, GRU ~0.5 ms). This enables real-time edge processing on drone companion computers or field gateway devices at ~14 frames per second.

Q30: How does Component 3 hand off its findings to Component 4 for autonomous intervention?

Component 3 serializes a 17-field validated Component4Output Pydantic JSON payload containing: severity score, severity category, progression trend, weekly rate of change, 7-day forecast score, and recommended agronomic action. Component 4 ingests this to schedule UAV precision spraying and calculate chemical tank mixes.